

Executive Summary: Expense Classification using NLP

1. Data Overview and EDA Findings

The dataset contains 268 expense entries from FY23 with columns: Year, Debit, Credit, Net, and Remarks. All Credit values are zero, making Net equal to Debit. The Remarks column contains free-text descriptions of expenses that form the basis for our NLP classification task.

Key EDA findings:

- * Debit amounts range from INR 300 to INR 12.7 million with high variance
- * Remarks text length varies from 5 to 200+ characters
- * Domain-specific vocabulary clearly differentiates expense categories
- * No explicit category labels exist; labels created using domain knowledge rules
- * Dataset is imbalanced: Equipment (~56%), Services (~27%), Material (~17%)
- * Services entries have longer text descriptions and higher debit amounts
- * Text features (word patterns) are more discriminative than numeric features alone

2. Data Cleaning

Steps performed: (1) Removed 1 empty row with no remarks text. (2) Converted numeric columns to proper types. (3) Preprocessed text by removing asset ID numbers, dates, PO numbers, and special characters. (4) Standardized text to lowercase for consistent feature extraction. These steps ensure the model focuses on meaningful textual content rather than noise from formatting inconsistencies.

3. Classification Approach Comparison

Approach 1: TF-IDF + Classical ML (Chosen)

Converts text to TF-IDF feature vectors capturing word/bigram importance, then trains classical classifiers (Logistic Regression, Random Forest, XGBoost, LightGBM, Extra Trees).

- * Advantages: Fast training, interpretable features, robust with small datasets (~268 samples), no GPU required, handles domain-specific vocabulary well
- * Disadvantages: Cannot capture deep semantic relationships, limited to surface-level word patterns, requires manual feature engineering

Approach 2: Transformer-based (BERT/DistilBERT)

Fine-tunes a pre-trained language model on the classification task, leveraging contextual embeddings learned from large corpora.

- * Advantages: Captures semantic meaning and context, handles synonyms and paraphrases, state-of-the-art NLP performance on large datasets
- * Disadvantages: Requires GPU infrastructure, prone to severe overfitting with only 268 samples, slow training, computationally expensive, difficult to interpret

Rationale for Approach 1: With only 268 samples, transformer models would overfit. The expense remarks contain strong keyword signals (e.g., "consultancy" for Services, "camera" for Equipment, "pipe" for Material) that TF-IDF captures effectively.

